

Assignments (Week 3)

2.4

2.6

2.19

2.21 (c) (d)

Tutorial Problems (Week 4)

- Basic Problems with Answers 2.3, 2.7, 2.13
- Basic Problems 2.24, 2.26

6) Linear and Nonlinear Systems

- Many systems are **nonlinear**.
 - ◆ e.g.: economic system with
input: fiscal and monetary policies, labor, resources, etc.
→ output: GDP, inflation, etc.
 - ◆ System behaviour is very unpredictable, because it is highly nonlinear.
- We will deal with only **linear** systems, which are good approximations of nonlinear systems in certain ranges.
 - ◆ e.g.: small-signal conductance of a nonlinear diode.
- Linear systems can be analyzed accurately.

Linearity

We have $x_1(t) \rightarrow y_1(t)$, and $x_2(t) \rightarrow y_2(t)$.

The system is linear, if:

- 1) Additivity property: $x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$
- 2) Scaling (or homogeneity) property:

$$a x_1(t) \rightarrow a y_1(t)$$

where a is a complex number

e.g.: $y(t) = 2 x(t)$

$$y(t) = x^2(t)$$

Linearity (cont.)

A (CT) system is linear if it has the superposition property:

If $x_1(t) \rightarrow y_1(t)$ and $x_2(t) \rightarrow y_2(t)$

then $ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$

- For linear systems, zero input $\rightarrow$ zero output

"Proof" $0 = 0 \cdot x[n] \rightarrow 0 \cdot y[n] = 0$

Question: Is the system $y = A + x$ linear?

- *Incrementally linear system!*

Linear system or not?

- **Steps**

- 1) Have $y_1(t)$ and $y_2(t)$ as output signals to $x_1(t)$ and $x_2(t)$
- 2) Have $y_3(t)$ as output signal to $x_3(t) = a x_1(t) + b x_2(t)$
- 3) Does $y_3(t)$ equal “ $a y_1(t) + b y_2(t)$ ”?

More examples on textbook

Read Example 1.17 ~ 1.20

Linearity (cont.)

- Superposition

If $x_k[n] \rightarrow y_k[n]$

Then

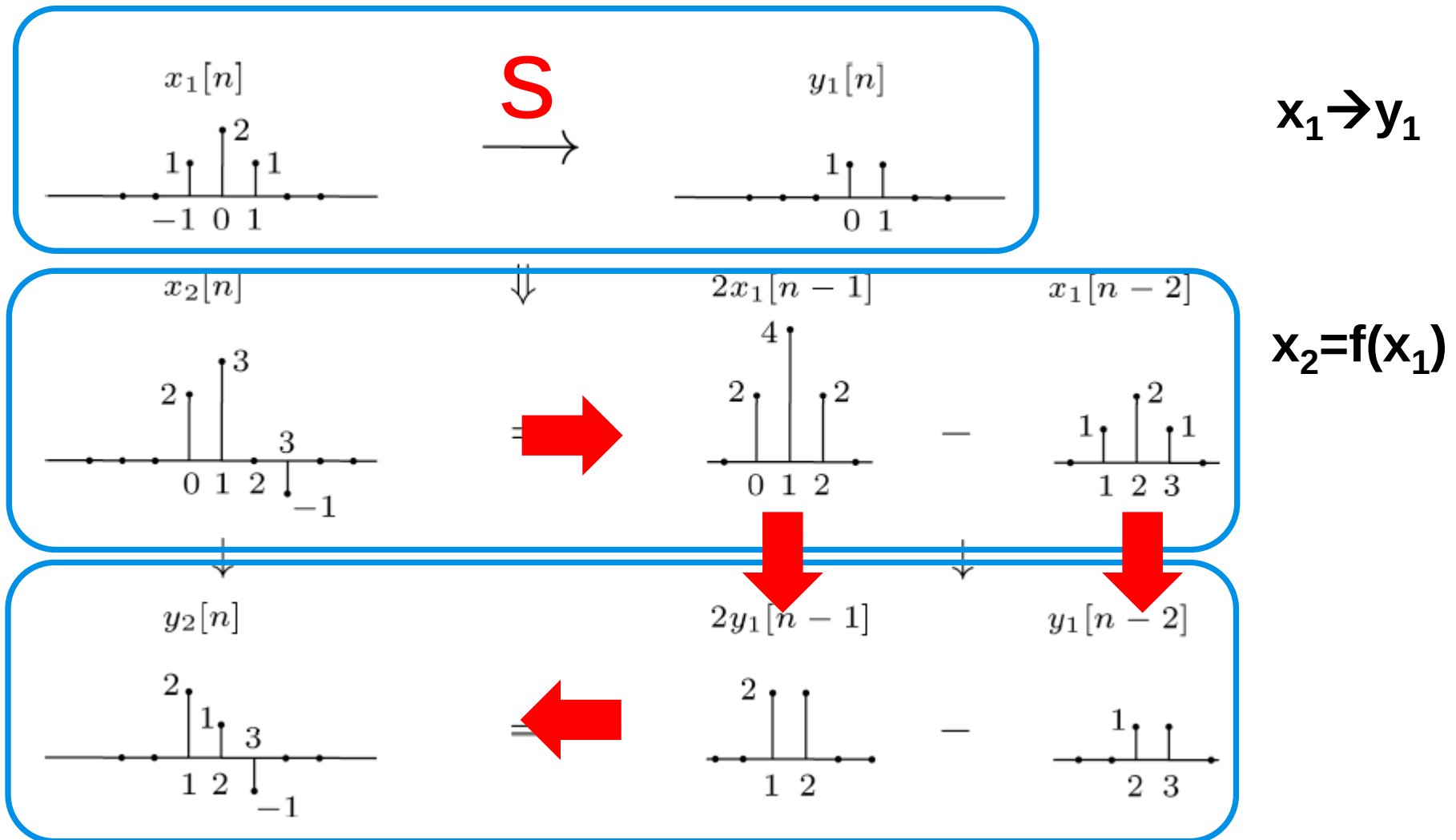
$$\sum_k a_k x_k[n] \rightarrow \sum_k a_k y_k[n]$$

- This property seems to be almost trivial now, but it is one of the most important ones

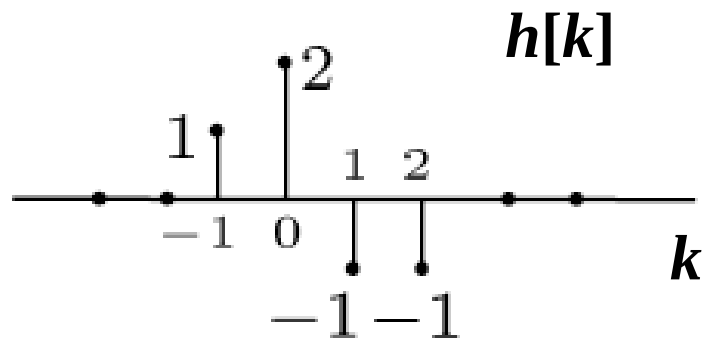
Linear Time-Invariant (LTI) Systems

- Focus of most of this course
 - ◆ Practical importance
 - ◆ The powerful analysis tools (*transformation*) associated with LTI systems
- A basic fact: If we know the response of an LTI system to **some** inputs, we actually know the response to **many** inputs.

Example: DT LTI System

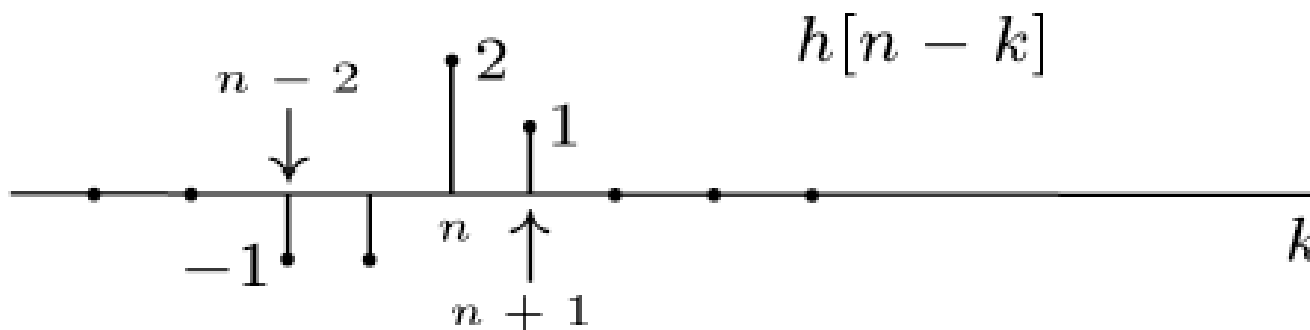


- Time-shift and flip



What is the plot for $h[n-k]$??
 n is a constant

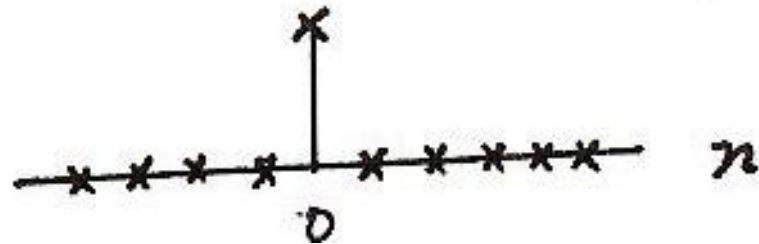
$$h[k] \xrightarrow{\text{Flip}} h[-k] \xrightarrow{\text{Slide}} h[n-k]$$



- Unit impulse function (unit sample function)

Discrete-time

$$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$



- We can use unit impulse function to represent any other different signals, or it is a building function (or basic signal).

System properties:

1. With memory or memoryless

$$y(n) = f(x(n))$$

2. Invertible (DIDO)

for a system $x \rightarrow y$, if $x_1 \neq x_2$, then $y_1 \neq y_2$

3. Causal

... up to that time n ...

4. Stable (BIBO)

either prove the system is stable, or find a specific counterexample

5. Time-invariant

$$\begin{array}{ll} \text{If} & x[n] \rightarrow y[n] \\ \text{then} & x[n - n_0] \rightarrow y[n - n_0] . \end{array}$$

6. Linear

A (CT) system is linear if it has the **superposition property**:

$$\text{If} \quad x_1(t) \rightarrow y_1(t) \quad \text{and} \quad x_2(t) \rightarrow y_2(t)$$

$$\text{then} \quad ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$$

Chapter 2

Linear Time-invariant (LTI) Systems

Exploiting Superposition and Time-Invariance

If we have $x_k[n] \rightarrow y_k[n]$, then

$$x[n] = \sum_k a_k x_k[n] \xrightarrow{\text{Linear System}} y[n] = \sum_k a_k y_k[n]$$

Question: Are there sets of “basic” signals $x_k[n]$ such that

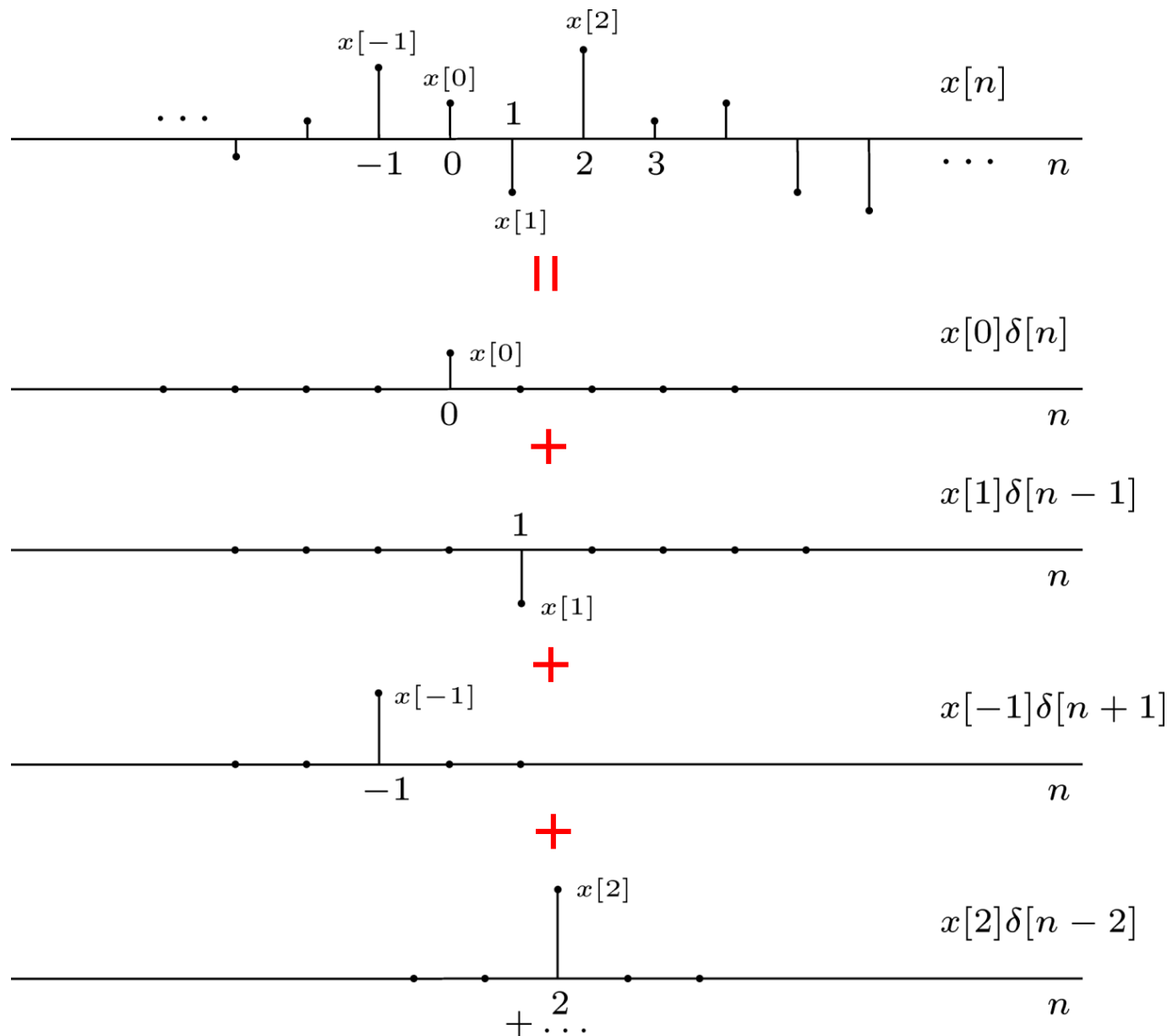
- We can represent *any* signals as linear combinations of these building block signals.
- The response of LTI Systems to these basic signals are both *simple* and *insightful*.

Fact: For LTI Systems (CT or DT) there are two natural choices for these building blocks

Focus for now: DT Shifted unit samples $\delta[n-n_o]$

Next time: CT Shifted unit impulses $\delta(t-t_o)$

Representation of DT Signals Using Unit Samples ¹⁵



That is ...

$$x[n] = \dots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \dots$$

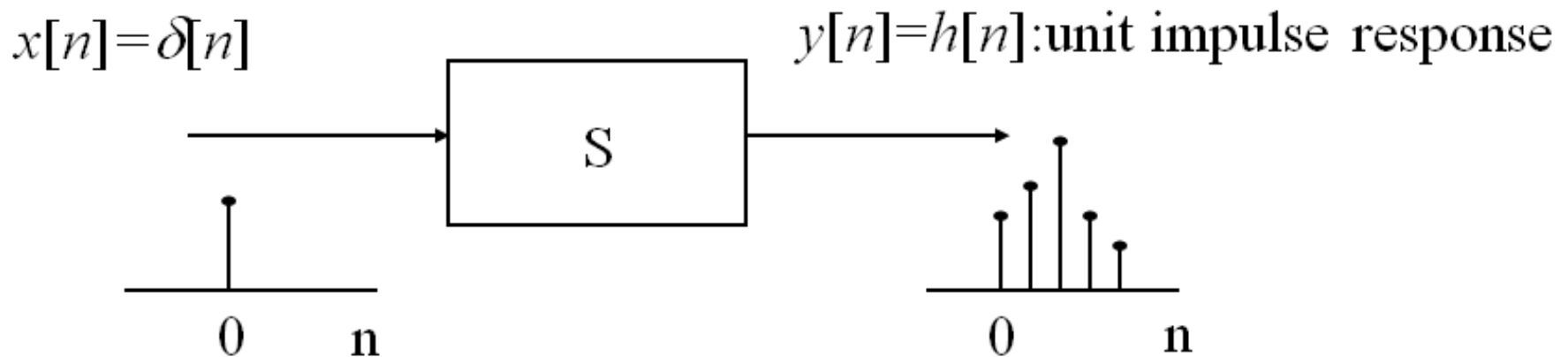
$$x[n] = \sum_{k=-\infty}^{+\infty} \underbrace{x[k]}_{\text{Coefficients}} \underbrace{\delta[n-k]}_{\text{Basic Signals}}$$

Important to note the “-” sign

- **Sifting property** of the unit impulse: looked at the index k , $\delta[n-k]$ is nonzero only at $k = n$, which “sifts” the value $x[n]$ out of the function $x[k]$.

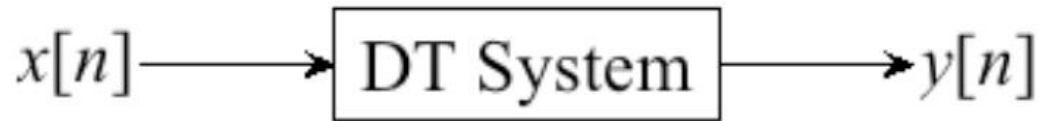
Unit Impulse Response (Unit Sample Response)

- Define the output for an **unit impulse input** as the **unit impulse response**



Example: $y[n] = x[n] + 2x[n-1] + 4x[n-2]$
 What is unit impulse response?

Response of DT LTI Systems



- Now suppose the system is **LTI**, and define the *unit impulse response* $h[n]$:

$$\delta[n] \longrightarrow h[n]$$



From **T**ime-**I**nvariance:

$$\delta[n - k] \longrightarrow h[n - k]$$

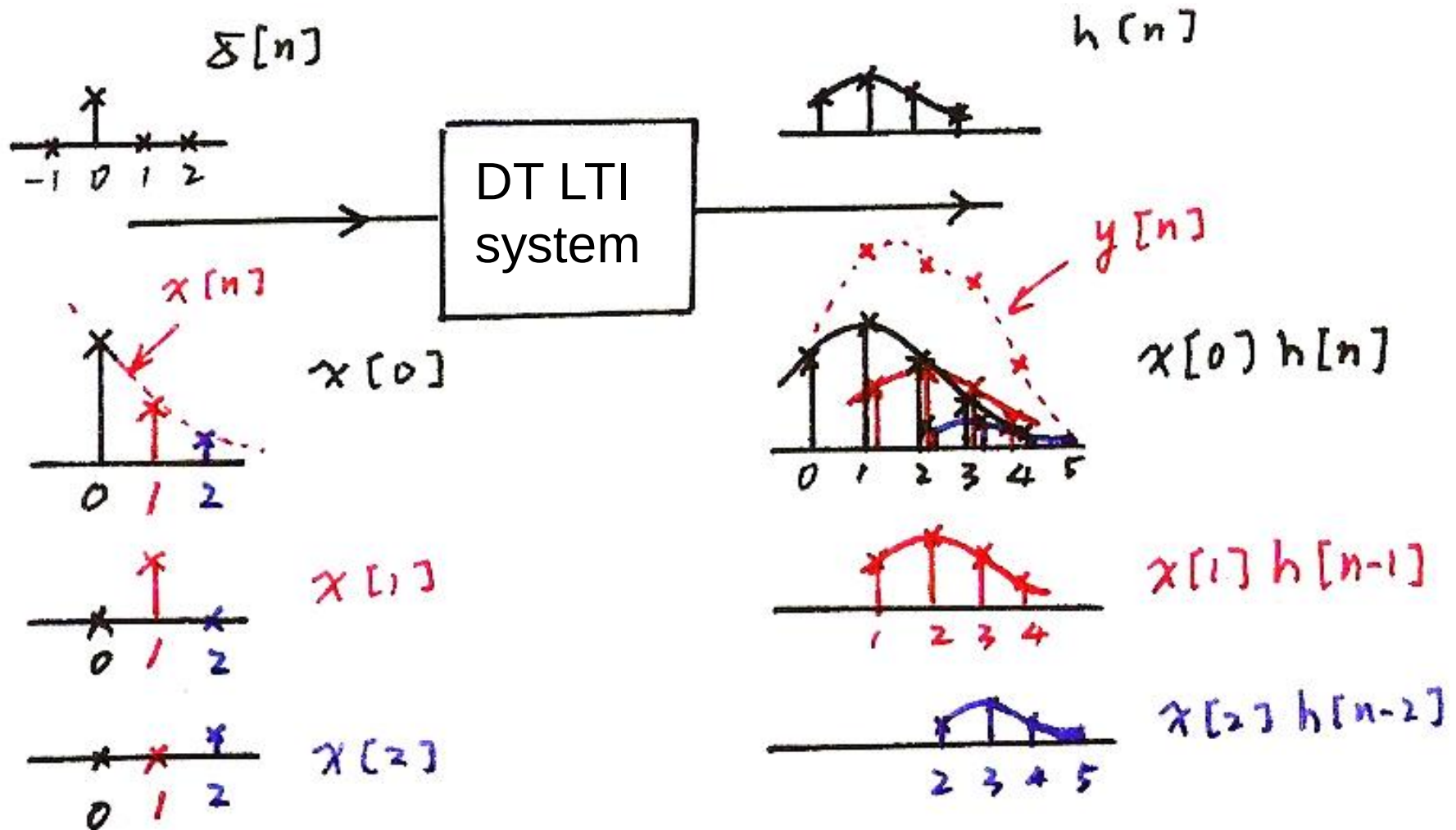
From **L**inearity:

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n - k] \longrightarrow y[n] = \underbrace{\sum_{k=-\infty}^{+\infty} x[k] h[n - k]}_{\text{convolution sum}} = x[n] * h[n]$$

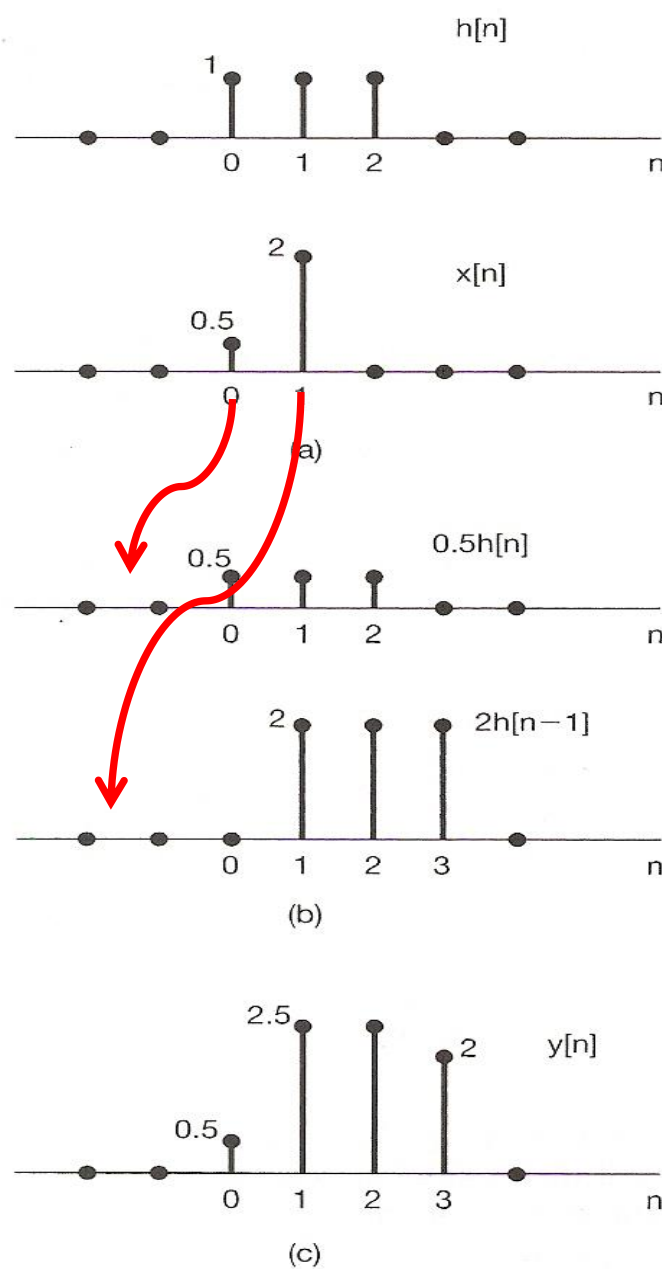
The output for an arbitrary input signal is the superposition of a series of “shifted, scaled unit impulse response”

Example

Input/Output Relation



Example



$$y[n] = 0.5h[n] + 2h[n-1]$$

Figure 2.3 (a) The impulse response $h[n]$ of an LTI system and an input $x[n]$ to the system; (b) the responses or “echoes,” $0.5h[n]$ and $2h[n-1]$, to the nonzero values of the input, namely, $x[0] = 0.5$ and $x[1] = 2$; (c) the overall response $y[n]$, which is the sum of the echoes in (b).

Hence a Very Important Property of LTI Systems:

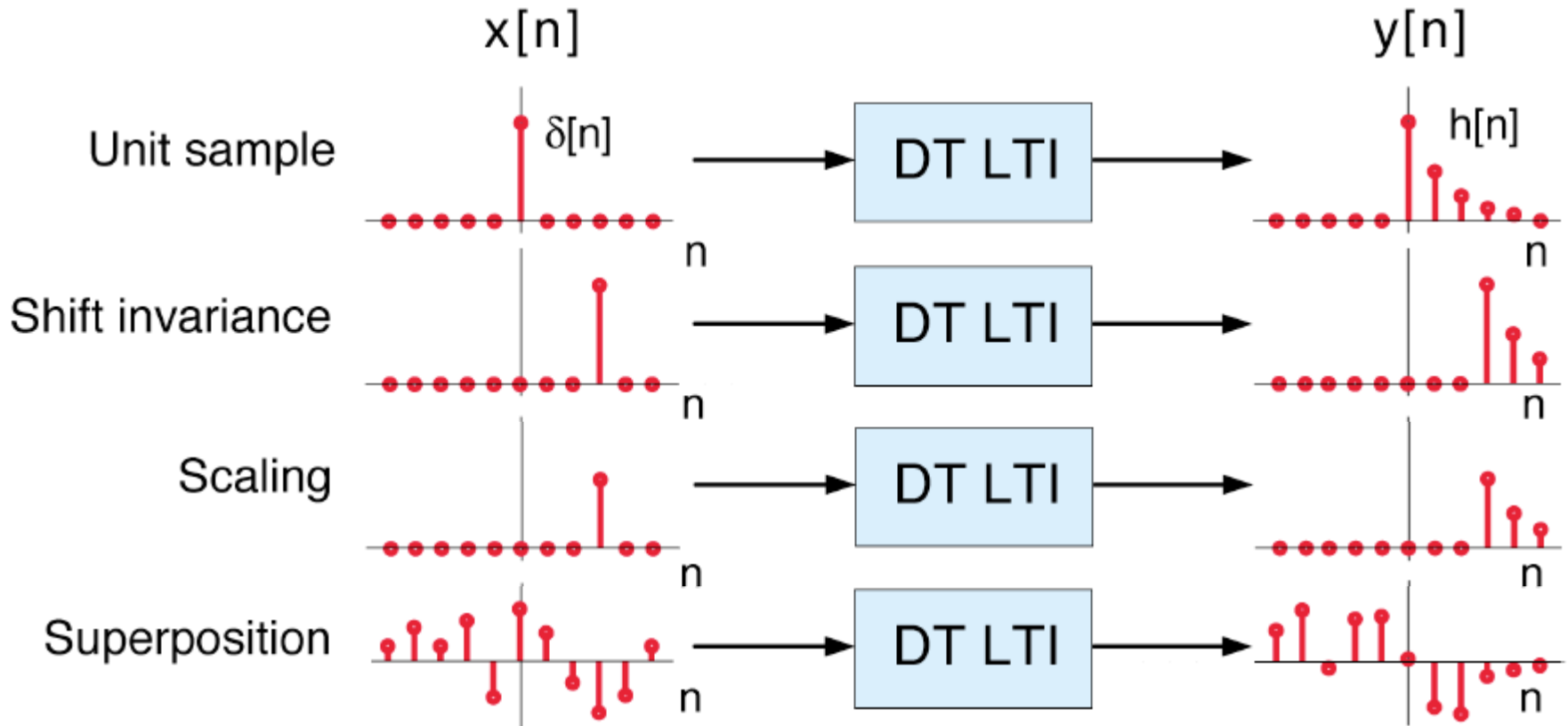
The output of *any* DT LTI System is a convolution of the input signal with the unit-sample response, *i.e.*

$$\begin{aligned}\text{Any DT LTI} &\longleftrightarrow y[n] = x[n] * h[n] \\ &= \sum_{k=-\infty}^{+\infty} x[k] h[n - k]\end{aligned}$$

As a result, any DT LTI Systems are *completely characterized* by its unit sample response

Graphic View of the Convolution Sum

Response of DT LTI systems



$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n-k] \xrightarrow{\text{DT LTI}} y[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n-k]$$

convolution sum

- A different way to visualize the convolution sum
 - looked at on the index k

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

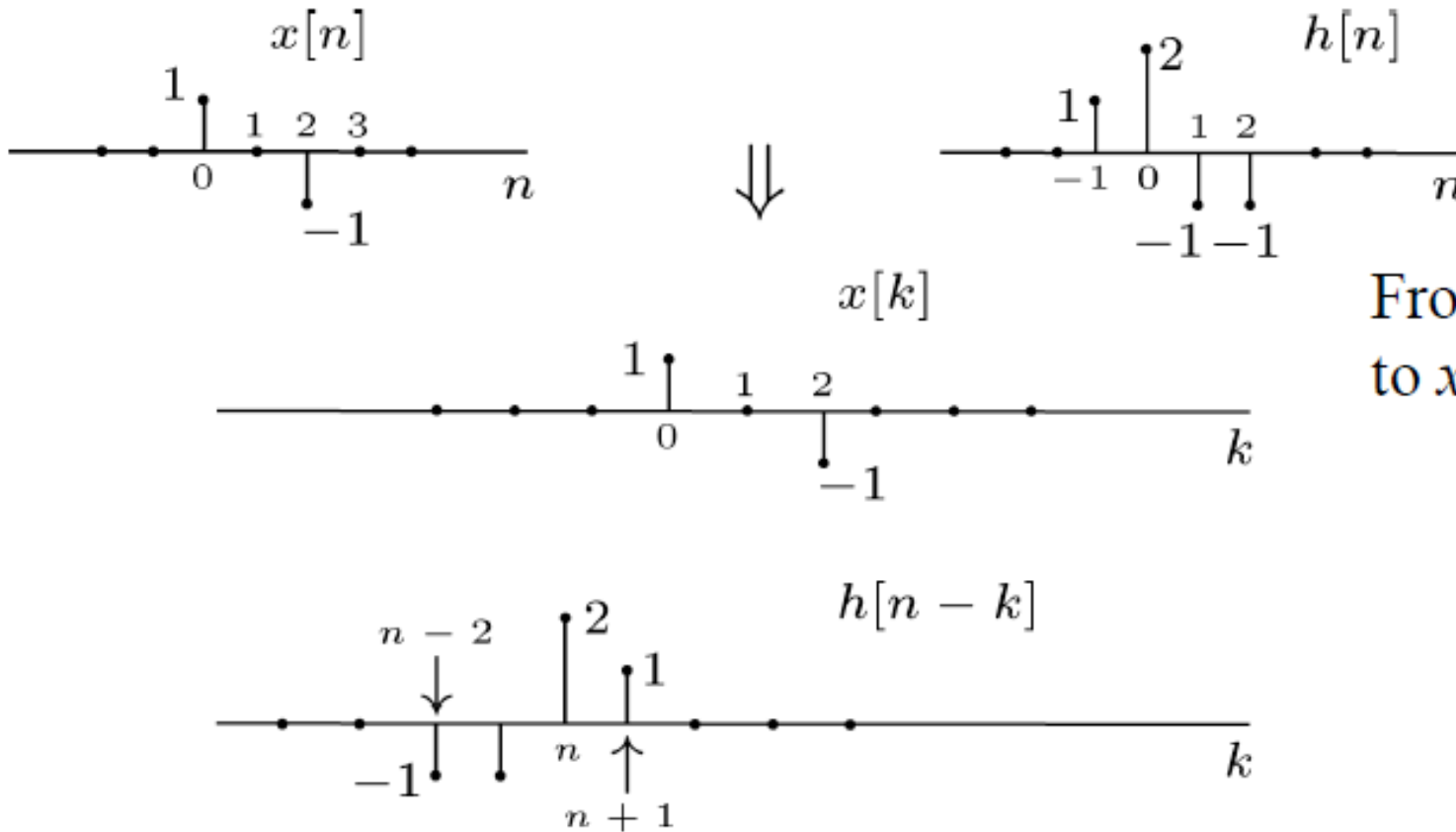
Contribution to the output signal at time n

input signal

flipped version of $h[k]$ located at $k = n$

- on the dummy index k , $h[k]$ is **flipped** over and **shifted** to $k=n$, **weighted** by $x[k]$, and **summed** to produce an output sample $y[n]$ at time n

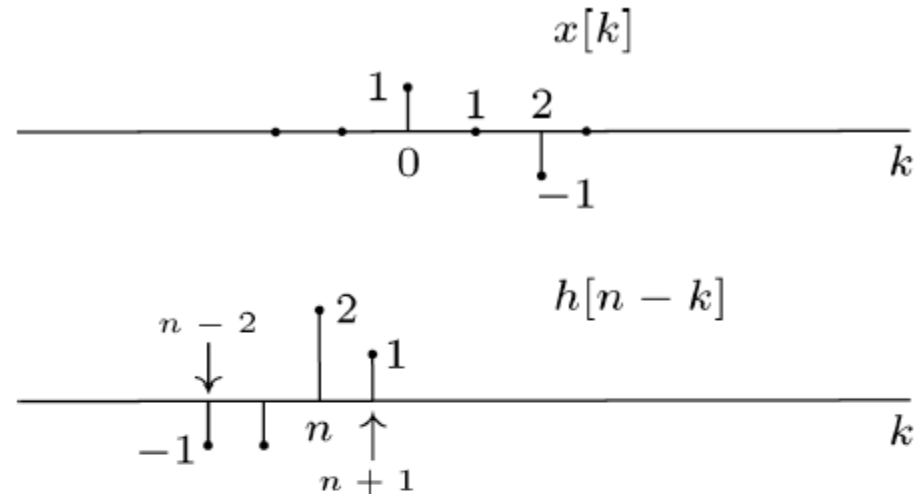
Example: Flip $h[n]$



From $x[n]$ and $h[n]$
to $x[k]$ and $h[n-k]$

Calculating Successive Values: Shift, Multiply, Sum

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$



$$y[n] = 0 \quad \text{for } n <$$

$$y[-1] =$$

$$y[0] =$$

$$y[1] =$$

$$y[2] =$$

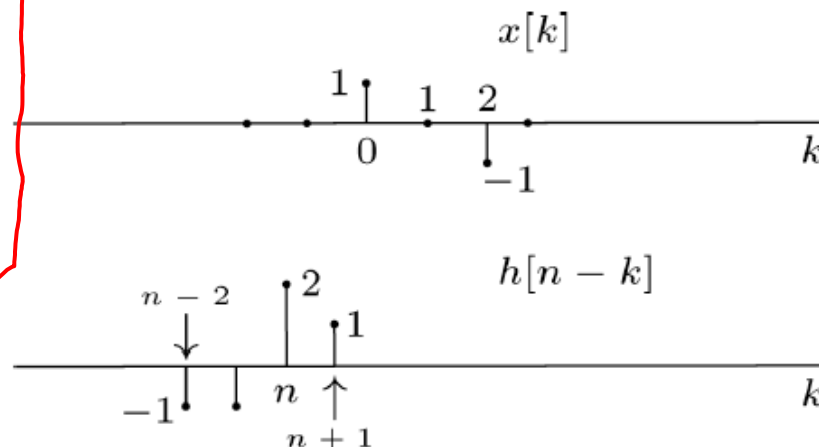
$$y[3] =$$

$$y[4] =$$

$$y[n] = 0 \quad \text{for } n >$$

Calculating Successive Values: **Shift, Multiply, Sum**

$$\begin{bmatrix} -1 & -1 & 2 & 1 & 0 & 0 & 0 & \dots \\ 0 & -1 & -1 & 2 & 1 & 0 & 0 & \dots \\ 0 & 0 & -1 & -1 & 2 & 1 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} x[-3] \\ x[-2] \\ x[-1] \\ x[0] \\ x[1] \\ \vdots \\ x[5] \end{bmatrix} = \begin{bmatrix} y[-1] \\ y[0] \\ y[1] \\ \vdots \\ y[4] \end{bmatrix}$$



If $\{-1, -1, 2, 1\}$ are unknown, $y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$

then we have 4 unknown variables and 6 equations. — Solvable!

$$\begin{aligned} y[n] &= 0 && \text{for } n < -3 \\ y[-1] &= (-1)x[-3] + (-1)x[-2] + 2x[-1] + 1x[0] \\ y[0] &= (-1)x[-2] + (-1)x[-1] + 2x[0] + 1x[1] \\ y[1] &= (-1)x[-1] + (-1)x[0] + 2x[1] + 1x[2] \\ y[2] &= \vdots \\ y[3] &= \vdots \\ y[4] &= (-1)x[2] + (-1)x[3] + 2x[4] + 1x[5] \\ y[n] &= 0 && \text{for } n > 5 \end{aligned}$$

$$y[n] = \sum_{k=-N}^N x[k]h[n-k]$$

$$\mathbf{y} = \mathbf{H}\mathbf{x}$$

$$\begin{bmatrix} y[-N] \\ y[-N+1] \\ \vdots \\ y[0] \\ \vdots \\ y[N-1] \\ y[N] \end{bmatrix} = \begin{bmatrix} h[0] & h[-1] & \dots & h[-N] & \dots & h[-2N+1] & h[-2N] \\ h[1] & h[0] & \dots & h[-N+1] & \dots & h[-2N+2] & h[-2N+1] \\ \vdots & \vdots & \dots & \vdots & \dots & \vdots & \vdots \\ h[N] & h[N-1] & \dots & h[0] & \dots & h[-N+1] & h[-N] \\ \vdots & \vdots & \dots & \vdots & \dots & \vdots & \vdots \\ h[2N-1] & h[2N-2] & \dots & h[N-1] & \dots & h[0] & h[-1] \\ h[2N] & h[2N-1] & \dots & h[N] & \dots & h[1] & h[0] \end{bmatrix} \begin{bmatrix} x[-N] \\ x[-N+1] \\ \vdots \\ x[0] \\ \vdots \\ x[N-1] \\ x[N] \end{bmatrix}$$



Toeplitz matrix

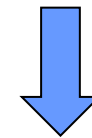
$$H_{i,j} = H_{i+1,j+1} = h_{i-j}$$

Convolution operation procedure:

$$h[k] \xrightarrow{\text{Flip}} h[-k] \xrightarrow{\text{Slide}} h[n-k] \xrightarrow{\text{Multiply}} x[k]h[n-k]$$

F S M S

$$\xrightarrow{\text{Sum}} \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$



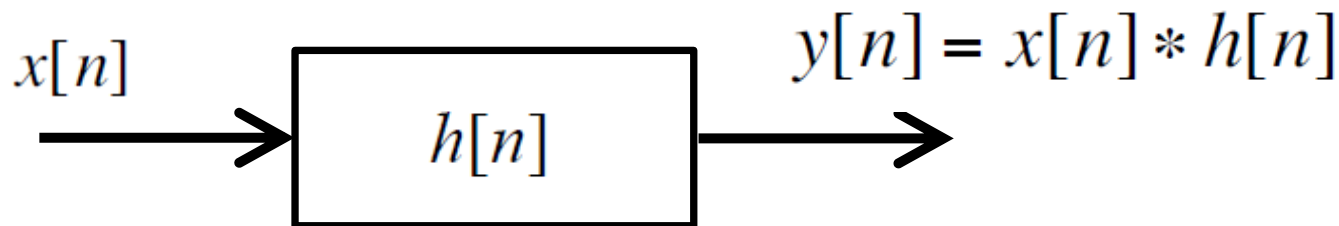
$y[n]$

$$\text{Any DT LTI} \iff y[n] = x[n] * h[n]$$
$$= \sum_{k=-\infty}^{+\infty} x[k] h[n - k]$$

$x[n]$: input

$y[n]$: output

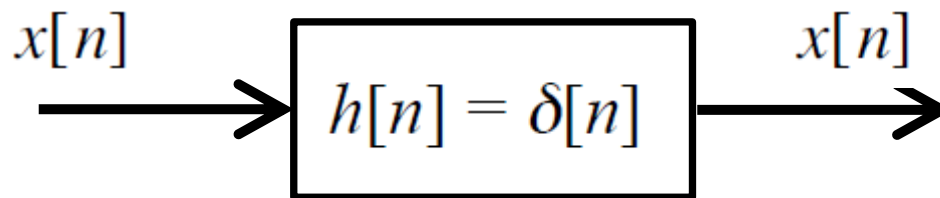
$h[n]$: impulse response of the system



Examples of Convolution and DT LTI Systems

Ex. #1: $h[n] = \delta[n]$

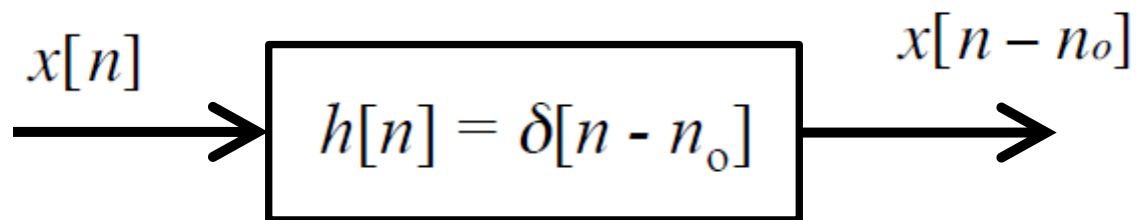
$$\begin{aligned} y[n] &= x[n] * \delta[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \\ &= x[n] \quad \text{— An Identity system} \end{aligned}$$



- sifting property, i.e., convolution sum (or integral) with a unit impulse function gives the original signal

Ex. #2: $h[n] = \delta[n - n_o]$

$$\begin{aligned} y[n] &= x[n] * \delta[n - n_o] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - n_o - k] \\ &= x[n - n_o] \quad \text{— A Shift} \end{aligned}$$



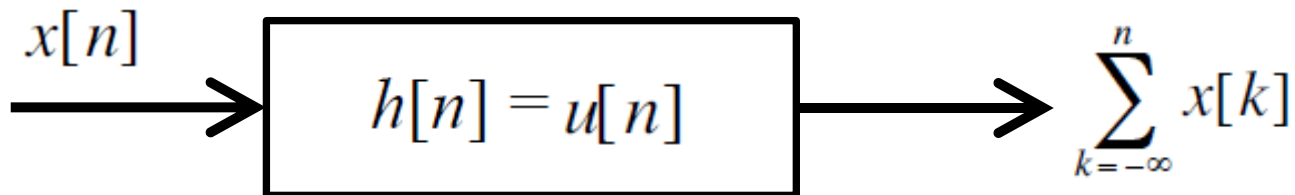
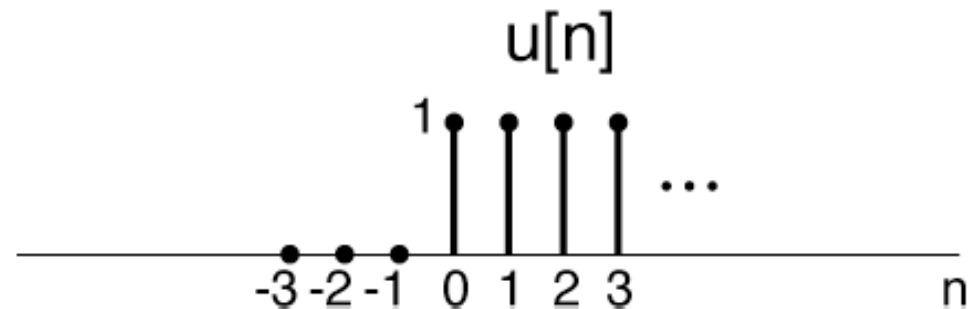
Ex. #3 $y[n] = \sum_{k=-\infty}^n x[k]$ – An accumulator

Unit Sample response

$$h[n] = \sum_{k=-\infty}^n \delta[k] = u[n]$$

↓

$$x[n] * u[n] = \sum_{k=-\infty}^n x[k]$$



Ex. #4 (Example 2.3)

$$0 < \alpha < 1$$

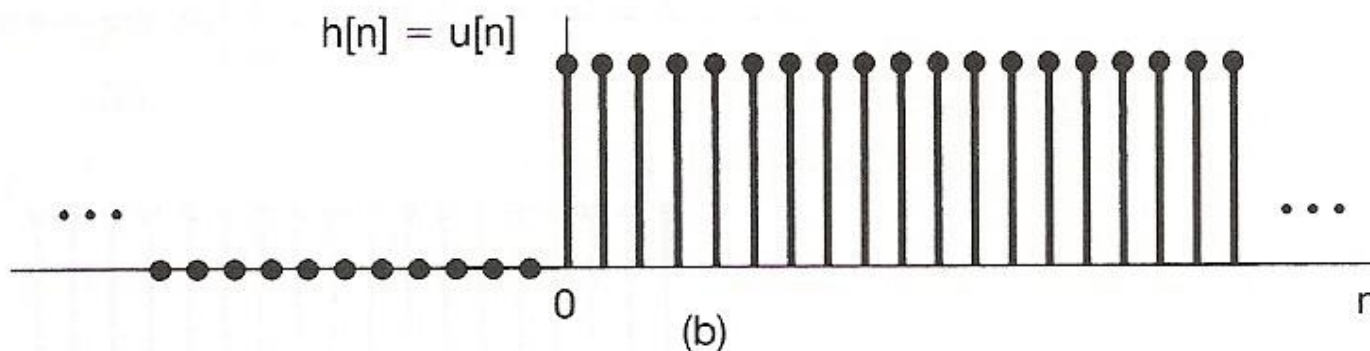
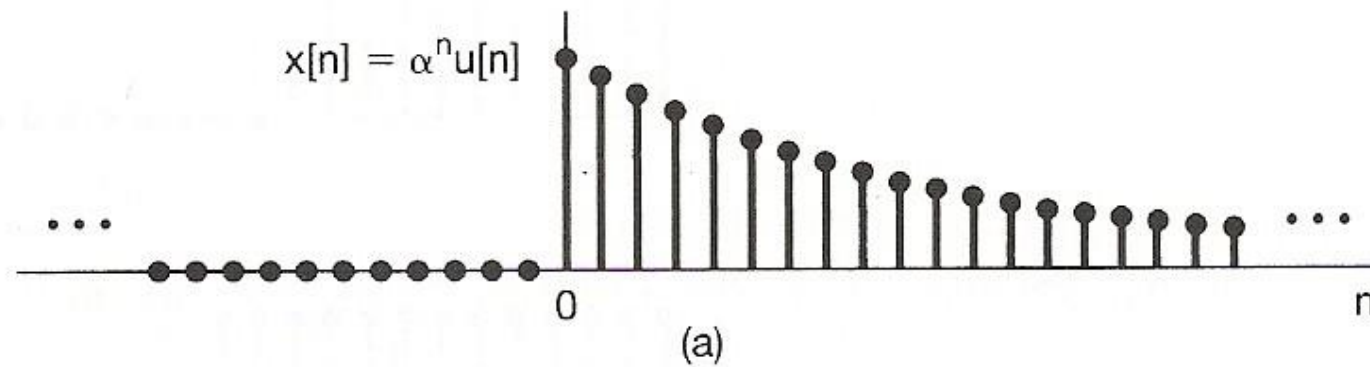


Figure 2.5 The signals $x[n]$ and $h[n]$ in Example 2.3.

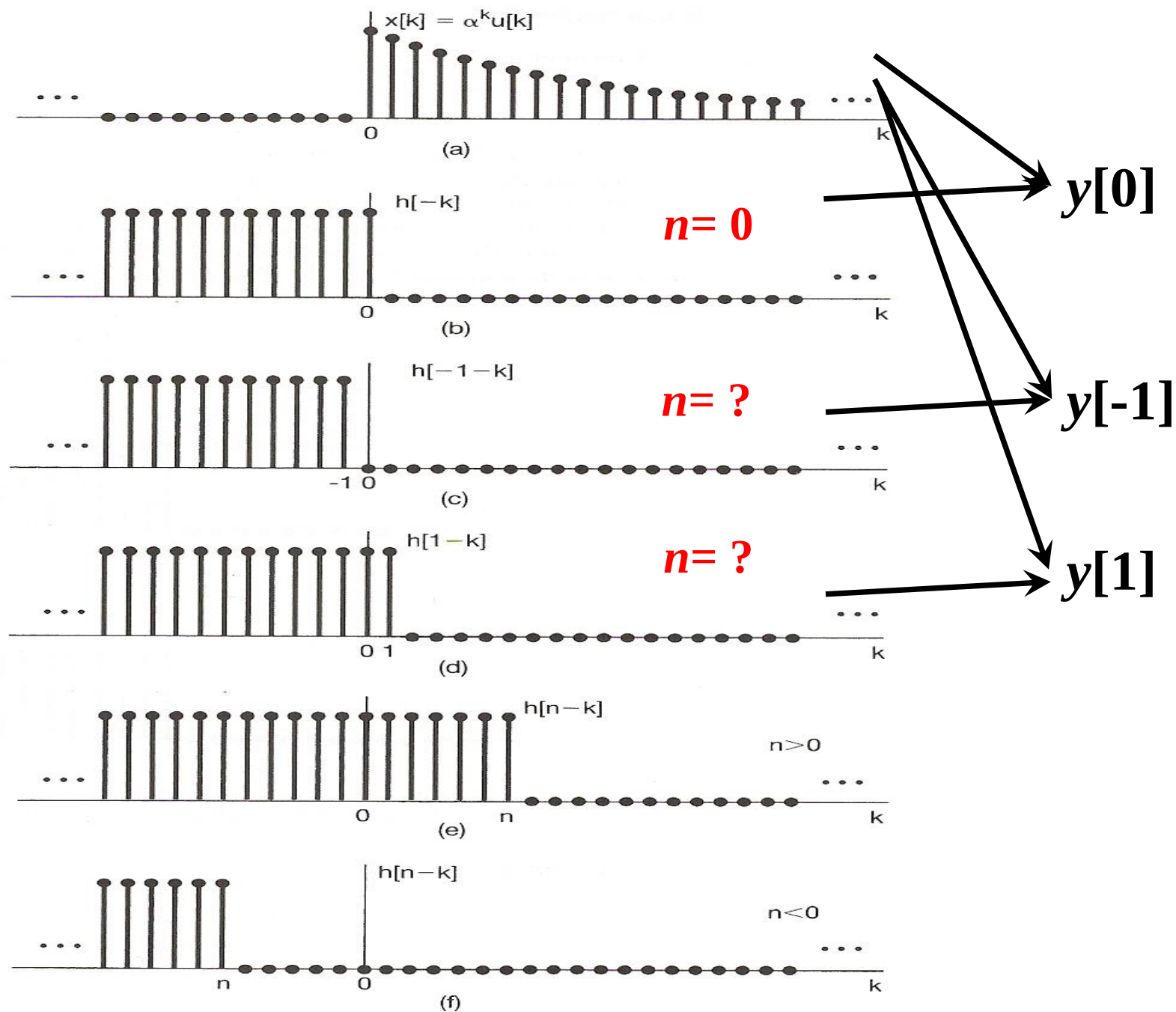


Figure 2.6 Graphical interpretation of the calculation of the convolution sum for Example 2.3.

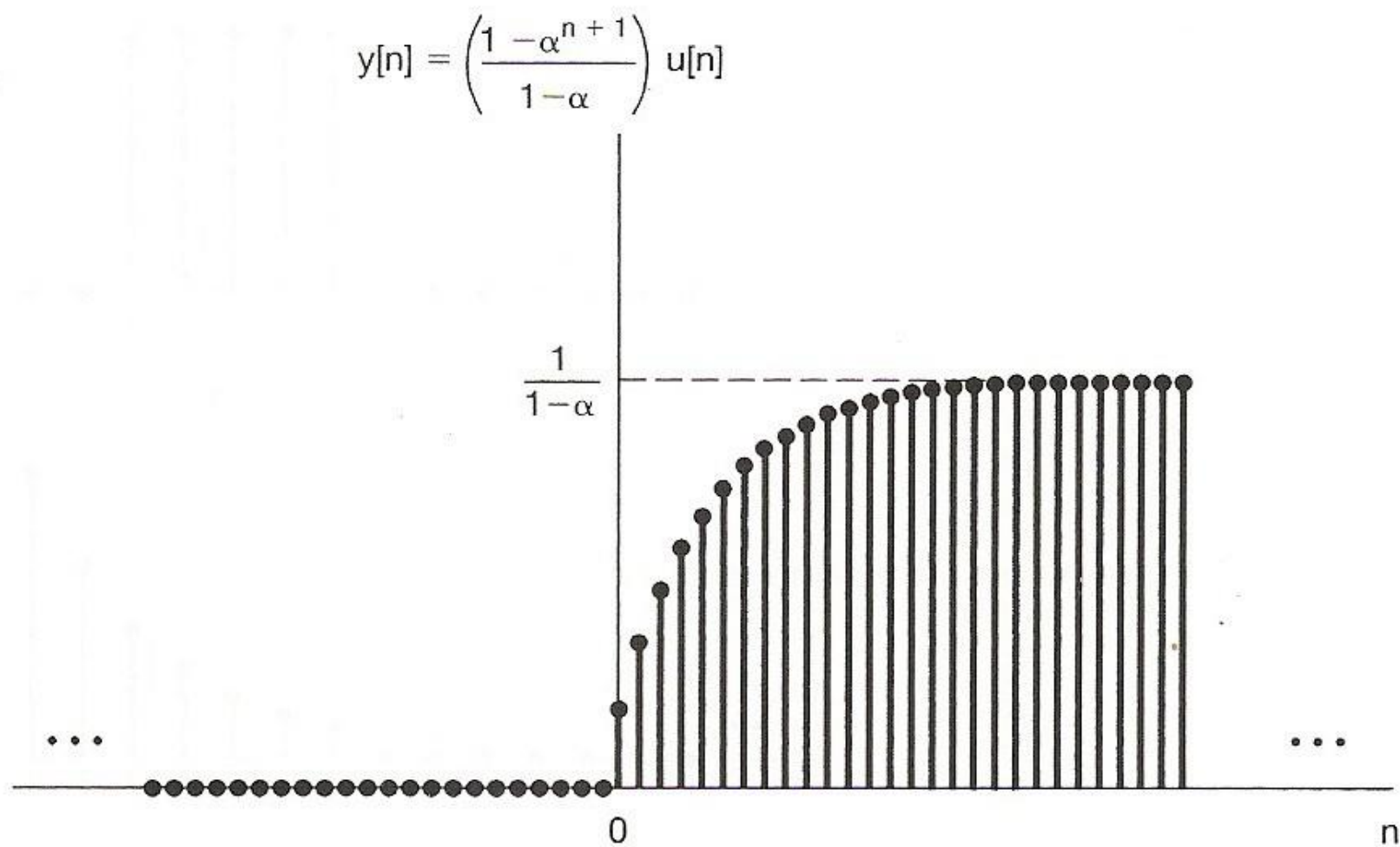


Figure 2.7 Output for Example 2.3.

Characteristics of an LTI system are completely determined by its impulse response.

- *What if the system is nonlinear?*

Consider a discrete-time system with unit impulse response

$$h[n] = \begin{cases} 1 & n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

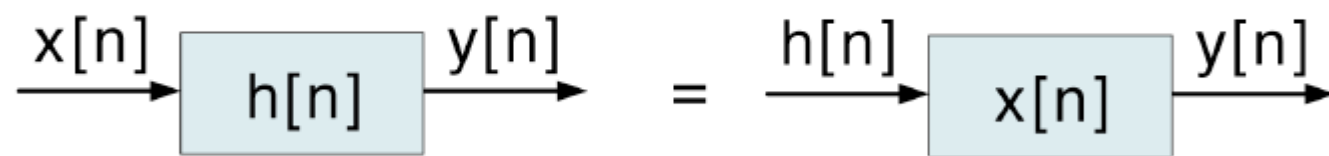
If the system is LTI, the input/output relationship is

$$y[n] = x[n] + x[n - 1].$$

On the other hand, there are *many* nonlinear systems with the same response to the input $\delta[n]$.

$$\begin{aligned} y[n] &= (x[n] + x[n - 1])^2, \\ y[n] &= \max(x[n], x[n - 1]). \end{aligned}$$

The Commutative Property of Convolution

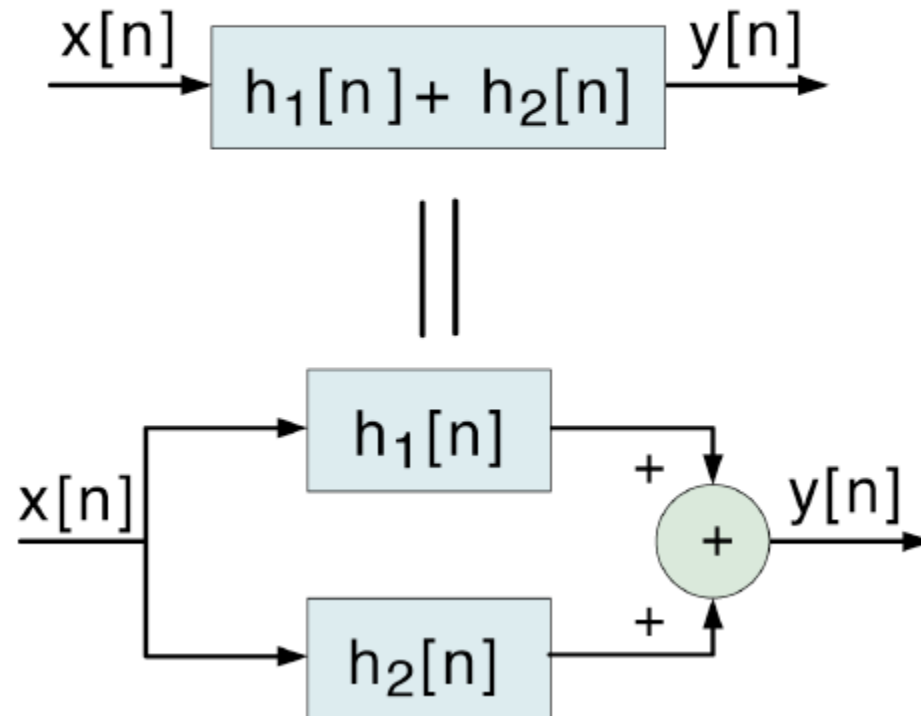
$$y[n] = x[n] * h[n] = h[n] * x[n]$$


The diagram illustrates the commutative property of convolution. It shows two equivalent block diagrams separated by an equals sign. On the left, an input signal $x[n]$ enters a block labeled $h[n]$, and the output is $y[n]$. On the right, the input signal is $h[n]$ entering a block labeled $x[n]$, with the same output $y[n]$.

The Distributive Property of Convolution

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

Interpretation



The Associative Property of Convolution

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

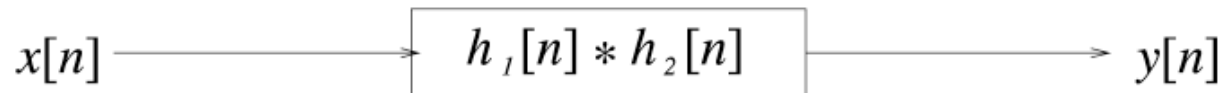
(Commutativity) ||

$$x[n] * (h_2[n] * h_1[n]) = (x[n] * h_2[n]) * h_1[n]$$

Implication (Very special to LTI Systems)



||



||



Properties of Convolution

Combining the Commutative property,

$$y[n] = x[n] * h[n] = h[n] * x[n]$$

Distributive property,

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

and Associative property,

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

symbolically, we can treat “*” as a “×”. Easy, piece of cake!

The hard part is the actual calculation of the convolution.

Flip → Slide → Multiply → Sum.

Soon we will develop a clever way (*transformation*) to perform “×” instead of “*” operation.

Some Useful Properties of LTI Systems

1) Causality $\Leftrightarrow h[n] = 0$ for all $n < 0$

2) Stability $\Leftrightarrow \sum_{k=-\infty}^{+\infty} |h[k]| < \infty$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

BIBO — Bounded Input $\Rightarrow$ Bounded Output

→ Sufficient condition: For $|x[n]| \leq x_{\max} < \infty$.

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} x[k]h[n-k] \right| \leq x_{\max} \left| \sum_{k=-\infty}^{\infty} h[n-k] \right| < \infty.$$

→ Necessary condition: If $\sum_{k=-\infty}^{\infty} |h[k]| = \infty$

Let $x[n] = h^*[-n]/|h[-n]|$, then $|x[n]| \equiv 1$ bounded

$$\text{But } y[0] = \sum_{k=-\infty}^{\infty} x[k]h[-k] = \sum_{k=-\infty}^{\infty} h^*[-k]h[-k]/|h[-k]| = \sum_{k=-\infty}^{\infty} |h[-k]| = \infty$$

● Memoryless / with Memory

– A linear, time-invariant, causal system is memoryless only

$$\text{if } h[n] = K\delta[n] \quad h(t) = K\delta(t)$$

$$y[n] = Kx[n] \quad y(t) = Kx(t)$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

if $k=1$ further, they are identity systems

$$y[n] = x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] = x[n] * \delta[n]$$

$$y(t) = x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau = x(t) * \delta(t)$$

Summary

Understand the following new concepts:

- 1. Use unit impulse function to represent any function**
- 2. Unit impulse response $h[n]$**
 - ◆ Given the system input/output equation, how to decide the unit impulse response?
- 3. Convolution, its properties, and calculation steps (FSMS)**
 - ◆ Understand the meaning of index 'k' and index 'n'
- 4. Decide LTI system property by using unit impulse response $h[n]$**